Lab 1 – Product Description

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1 Introduction

1.1 Background

Urban populations continue to grow as more individuals relocate to metropolitan areas for employment and education. This growth places significant strain on transportation systems, resulting in increased congestion, longer commute times, higher fuel consumption, and environmental impact. Studies show that urban overcrowding contributes directly to economic and environmental inefficiencies (Joyce, 2025; Lee, 2025).

Although navigation technologies are widely available, rapidly changing traffic conditions require more dynamic and localized solutions. The widespread use of smartphones creates an opportunity to leverage real time data and user participation to improve urban mobility.

1.2 Problem Statement

Current navigation applications provide routing assistance but often lack detailed, localized insight into temporary congestion, hazards, and community reported conditions. As a result, commuters frequently encounter delays despite using existing tools. There is a need for a system that integrates navigation with real time, community driven information to improve situational awareness and route efficiency.

1.3 Proposed Solution

Smart City is a mobile and web based application designed to improve urban navigation through real time routing and community engagement. The system integrates mapping services, user authentication, and a location based forum that allows users to share congestion updates and local conditions.

By combining navigation tools with user generated insights, the Real World Product (RWP) aims to reduce commute inefficiencies and enhance overall urban awareness.

1.4 Scope

This document describes the Smart City Real World Product and its CS411W prototype. The scope includes the system concept, major components, and primary features.

The prototype demonstrates core functionality such as navigation integration, user authentication, and community forum interaction. Advanced analytics, full-scale deployment, and enterprise-level scalability are outside the scope of the prototype and reserved for future development.

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2 Product Description

Smart City is an environmentally aware application designed to help users navigate urban environments more efficiently whilst encouraging community engagement via forum. The product provides real time navigation support, nearby location recommendations, and a public forum where users can share updates about congestion, hazards, or local conditions. The primary goal of Smart City is to improve daily mobility and city awareness by combining traditional navigation functionality with community generated insights.

By integrating mapping services with social interaction, Smart City differentiates itself from existing navigation tools. Rather than relying solely on static or algorithm-driven data, the product allows users to actively contribute information. This collaborative approach enables more accurate, timely, and relevant guidance for city residents. Ultimately, Smart City seeks to reduce commute times, improve user satisfaction, and strengthen connections between individuals and their urban surroundings.

2.1 Key Product Features and Capabilities

The Smart City product offers several key features that directly address the challenges of urban overcrowding. These include real-time routing designed to avoid congested areas, personalized recommendations for nearby locations, and a location based community forum. Users can view and contribute updates related to traffic conditions, obstacles, or neighborhood activity, enhancing situational awareness for others.

As well, the system provides a simple dashboard that presents insights related to city activity and user behavior. Together, these features create a platform that not only assists users in reaching their destinations more efficiently but also empowers them to participate in improving urban mobility for their community as a whole. The integration of navigation and community interaction represents a unique and innovative approach to solving congestion related issues.

2.2 Major Components (Hardware/Software)

Smart City is composed of several hardware and software components that support its functionality. The primary user interface is a React Native application deployed on Android and iOS, along with a web-based data platform. A Node.js backend manages application logic, while a MySQL database stores user data, forum content, and location-related information.

The system relies on external APIs to provide essential services, including Mapbox and OpenStreetMap for mapping and routing, and Firebase for user authentication and account management. These components work together to form a scalable architecture capable of supporting real time navigation, data processing, and community interaction.

3 Identification of Case Study

3.1 Case Study Overview

The Smart City prototype is primarily developed for urban residents who frequently navigate congested city environments. These users are key to understanding how the application can optimize navigation and provide valuable feedback on usability, navigation accuracy, and the overall user experience.

3.2 Target User Group

The initial case study group consists of university students and daily commuters. This demographic is ideal due to their daily reliance on navigation tools and exposure to urban congestion. Their consistent usage patterns provide useful data on app functionality and user interaction with the platform's core features.

3.3 Potential for Future Expansion

As the product evolves, the target audience may expand beyond university students and daily commuters. Future versions of Smart City could serve a broader range of users, including other city residents, visitors, and even urban planners. By analyzing usage patterns and community feedback, the Smart City app could help inform long-term urban planning decisions, supporting the development of smarter, more responsive cities.

4 Product Prototype Description

The CS411W prototype of Smart City represents a reduced but functional version of the full CS410 product. The prototype focuses on demonstrating core capabilities such as navigation, user authentication, and community interaction while limiting or simulating more complex features. Certain functionalities, including advanced analytics and real-time alerts, are partially implemented or modeled to reflect a more feasible goal within our given timeline.

Despite these reductions, the prototype effectively illustrates the intended behavior and value of the final product. By showcasing key interactions and workflows, the prototype validates design decisions and highlights how Smart City addresses the challenges listed in the introduction.

4.1 Prototype Architecture (Hardware/Software)

The prototype architecture mirrors the structure of the final Smart City system. It utilizes React Native for the user interface, a backend API for data handling, and external mapping services for navigation. The Major Functional Components Diagram illustrates how the presentation, application, and data layers interact to deliver a functional end product.

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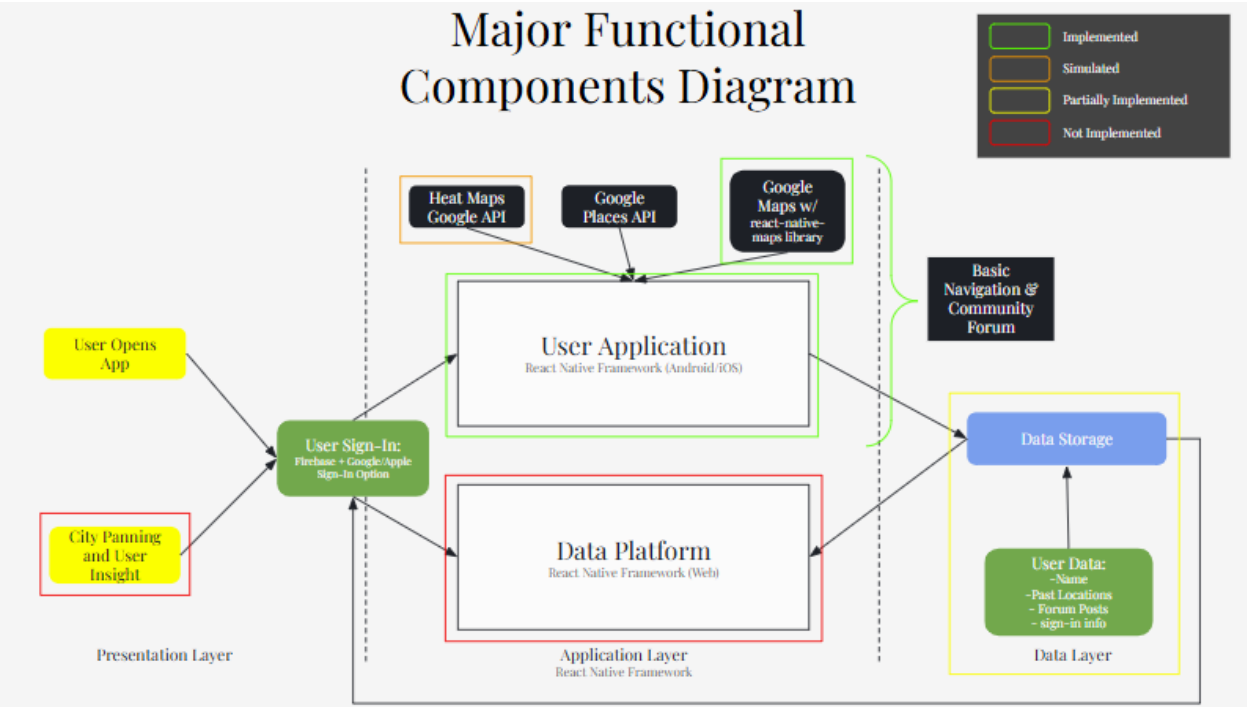
Figure 1



This architecture allows the prototype to demonstrate essential features while maintaining scalability. The design supports future expansion and integration of additional services as the product evolves into something greater.

4.2 Prototype Features and Capabilities

The prototype demonstrates basic navigation functionality, user account management, and a community forum. Some features, such as alerts and data analytics, are simulated or partially implemented due to complexity and time constraints of the semester. A comparison between the Real World Product (RWP) and the prototype highlights which features are fully implemented, partially implemented, or deferred.

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Table 1

Table of Comparison Between RWP and Prototype

Category	Feature	RWP	Prototype
Account Management	Login/Authentication	Fully Implemented	Fully Implemented
	Account Creation/Deletion	Fully Implemented	Fully Implemented
	Manage Account Access/Permissions	Fully Implemented	Fully Implemented
User Application	Access Community Forum	Fully Implemented	Fully Implemented
	Access Dynamic Queue System for Navigation	Fully Implemented	Partially Implemented
	Alerts and Notifications	Fully Implemented	Partially Implemented
	Add Parking Locations	Fully Implemented	Not Implemented

Data Platform	Central Information Hub	Fully Implemented	Partially Implemented
	Data Analytics and Insights	Fully Implemented	Not Implemented
	Visualize Data	Fully Implemented	Not Implemented
	Data Anonymity	Fully Implemented	Fully Implemented

These demonstrations are sufficient to show how the product mitigates key project risks and achieves its primary goals. The prototype provides evidence that the Smart City concept is both valuable and feasible.

4.3 Prototype Development Challenges

Several challenges are anticipated during prototype development. Such challenges include integrating mapping services with forum functionality, managing simulated data during demonstrations, and ensuring acceptable performance during map rendering. Additional challenges involve balancing feature scope with project timelines and maintaining a clean, usable interface.

Addressing these challenges through careful design and risk mitigation strengthens the overall prototype and helps with future development efforts.

5 Glossary

1. **RWP – Real World Product**

2. **Smart City — an innovative city that uses ICT and other means to improve quality of life and urban operations while advancing sustainability**

3. **Internet of Things (IoT) — systems that combine sensing, computing, communication, and actuation to connect physical/cyber objects**

4. **API (Application Programming Interface) — a set of features and rules that lets software interact with other software (a "contract" between programs)**

5. **Crowdsourced Reporting — gathering information by soliciting contributions from a large group of people, typically via the internet**

6. **Privacy-by-Design ("Privacy-First Design") — building systems so that, by design and by default, only the personal data necessary for a purpose are processed**

7. **Open Data (Open Data Access) — data anyone can freely use, re-use, and redistribute, with at most attribution/share-alike requirements**

8. **Predictive AI / AI-Powered Forecasting — using data, statistical algorithms, and ML to estimate the likelihood of future outcomes**

9. **Anomaly Detection — identifying data patterns that deviate from expected behavior**

10. **Access Control — controls that enforce authorized access to information and system resources in line with policy**

11. **Citizen Feedback Loop — a two-way process where institutions act on and report back about citizen input to sustain engagement and trust**

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